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STA506
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Homework 5

Question 1: Derive gradient (first order partial derivative) of likelihood function.

a.

Step 1: Write the probability density function

$$f(x|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} \quad \text{for } x > 0$$

Step 2: Write the likelihood function

$$L(\alpha, \beta) = \prod_{i=1}^n f(x_i|\alpha, \beta)$$

****Substitute the PDF****

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} e^{-x_i/\beta}$$

Step 3: Take the natural log to get the log-likelihood

$$\ell(\alpha, \beta) = \log L(\alpha, \beta)$$

$$\ell(\alpha, \beta) = -n \log \Gamma(\alpha) - n \alpha \log \beta + (\alpha - 1) \sum_{i=1}^n \log x_i - \frac{1}{\beta} \sum_{i=1}^n x_i$$

b.

$$\ell(\alpha, \beta) = -n \log \Gamma(\alpha) - n \alpha \log \beta + (\alpha - 1) \sum_{i=1}^n \log x_i - \frac{1}{\beta} \sum_{i=1}^n x_i$$

Step 1: Differentiate with respect to α

$$\frac{\partial \ell}{\partial \alpha} = \frac{\partial}{\partial \alpha} [-n \log \Gamma(\alpha)] + \frac{\partial}{\partial \alpha} [-n \alpha \log \beta] + \frac{\partial}{\partial \alpha} \left[(\alpha - 1) \sum_{i=1}^n \log x_i \right] + \frac{\partial}{\partial \alpha} \left[-\frac{1}{\beta} \sum_{i=1}^n x_i \right]$$

****Evaluate each term****

**Use $\frac{d}{d\alpha} \log \Gamma(\alpha) = \psi(\alpha)$ **

$$\frac{\partial}{\partial \alpha} [-n \log \Gamma(\alpha)] = -n \psi(\alpha)$$

$$\frac{\partial}{\partial \alpha} [-n \alpha \log \beta] = -n \log \beta$$

$$\frac{\partial}{\partial \alpha} \left[(\alpha - 1) \sum_{i=1}^n \log x_i \right] = \sum_{i=1}^n \log x_i$$

$$\frac{\partial}{\partial \alpha} \left[-\frac{1}{\beta} \sum_{i=1}^n x_i \right] = 0$$

$$\frac{\partial \ell}{\partial \alpha} = -n \psi(\alpha) - n \log \beta + \sum_{i=1}^n \log x_i$$

Step 2: Differentiate with respect to β

$$\frac{\partial \ell}{\partial \beta} = \frac{\partial}{\partial \beta} [-n \log \Gamma(\alpha)] + \frac{\partial}{\partial \beta} [-n \alpha \log \beta] + \frac{\partial}{\partial \beta} \left[(\alpha - 1) \sum_{i=1}^n \log x_i \right] + \frac{\partial}{\partial \beta} \left[-\frac{1}{\beta} \sum_{i=1}^n x_i \right]$$

Evaluate each term

$$\frac{\partial}{\partial \beta} [-n \log \Gamma(\alpha)] = 0$$

$$\frac{\partial}{\partial \beta} [-n \alpha \log \beta] = -n \alpha \cdot \frac{1}{\beta} = -\frac{n \alpha}{\beta}$$

$$\frac{\partial}{\partial \beta} \left[(\alpha - 1) \sum_{i=1}^n \log x_i \right] = 0$$

**Use $\frac{d}{d\beta} \left(\frac{1}{\beta} \right) = -\frac{1}{\beta^2}$ **

$$\frac{\partial}{\partial \beta} \left[-\frac{1}{\beta} \sum_{i=1}^n x_i \right] = - \left(-\frac{1}{\beta^2} \right) \sum_{i=1}^n x_i = \frac{1}{\beta^2} \sum_{i=1}^n x_i$$

$$\frac{\partial \ell}{\partial \beta} = -\frac{n \alpha}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n x_i$$

Score function for α

$$\frac{\partial \ell}{\partial \alpha} = -n\psi(\alpha) - n\log\beta + \sum_{i=1}^n \log x_i$$

Score function for β

$$\frac{\partial \ell}{\partial \beta} = -\frac{n\alpha}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n x_i$$

Question 2: Birth data set

a.

```

{r}
x <- c(2.1, 3.4, 4.25, 5.6, 6.4, 7.3, 8.5, 8.75, 8.9, 9.5, 9.75, 10, 10.4, 10.4, 16,
19,
      4, 4.1, 5, 5.5, 5.7, 6.5, 7.25, 7.3, 7.5, 8.2, 8.5, 9.75, 11, 11.2, 15, 16.5,
2.6,
      3.6, 3.6, 6.4, 6.8, 7.5, 7.5, 8.25, 8.5, 10.4, 10.75, 14.25, 14.5, 1.5, 4.7,
4.7,
      7.2, 7.25, 8.1, 8.5, 9.2, 9.5, 10.7, 11.5, 2.5, 2.5, 3.4, 4.2, 5.9, 6.25, 7.3,
7.5,
      7.8, 8.3, 8.3, 10.25, 12.9, 14.3, 4, 4, 5.25, 6.1, 6.5, 6.9, 7, 8.45, 9.25,
10.1,
      10.2, 12.75, 14.6, 2, 2.7, 2.75, 3.4, 4.2, 4.3, 4.9, 6.25, 7, 9, 9.25, 10.7)
n <- length(x)

```

```

{r}
negloglik <- function(par, x) {
  alpha <- par[1]
  beta <- par[2]
  n <- length(x)

  if (alpha <= 0 || beta <= 0) return(Inf)

  ll <- -n * lgamma(alpha) -
    n * alpha * log(beta) +
    (alpha - 1) * sum(log(x)) -
    (1 / beta) * sum(x)

  -ll
}

```

```

{r}
neggrad <- function(par, x) {
  alpha <- par[1]
  beta <- par[2]
  n <- length(x)

  if (alpha <= 0 || beta <= 0) return(c(NA, NA))

  dlda <- -n * digamma(alpha) - n * log(beta) + sum(log(x))
  dldb <- -(n * alpha / beta) + (sum(x) / beta^2)

  -c(dlda, dldb)
}

```

```

```{r}
m <- mean(x)
s2 <- var(x)

start <- c(alpha = m^2 / s2,|
 beta = s2 / m)
```

```

```

```{r}
fit <- optim(par = start,
 fn = negloglik,
 gr = neggrad,
 x = x,
 method = "L-BFGS-B",
 lower = c(1e-8, 1e-8))

fit$par
fit$convergence
```

```

```

> fit$par
      alpha      beta
4.387855 1.760122
> fit$convergence
[1] 0
>

```

Using the `optim()` to maximum likelihood estimates are $\hat{\alpha} = 4.387855$ and $\hat{\beta} = 1.760122$. The optimization converged successfully since `convergence = 0`.

b.

```

```{r}
xbar <- mean(x)
s2 <- var(x)

alpha_tilde <- xbar^2 / s2
beta_tilde <- s2 / xbar

alpha_tilde
beta_tilde
```

```

```

> alpha_tilde
[1] 4.685073
> beta_tilde
[1] 1.64846

```

Using the method of moments, the parameter estimate are $\tilde{\alpha} = 4.685073$ and $\tilde{\beta} = 1.64846$. The sample mean and sample variance was used to compute.

c.

The method of moments is a technique for deriving estimators by equating population moments with sample moments. The population moments of the Gamma distribution are the mean and variance given by $E[X] = \alpha\beta$ and $Var(X) = \alpha\beta^2$. The sample moments are given by the sample

mean $\bar{x}$ and sample variance s^2 . When we equate the population and sample moments, we get the estimators $\tilde{\alpha} = \bar{x}^2/s^2$ and $\tilde{\beta} = s^2/\bar{x}$. The method of moments is very simple to implement. It does not however use any of the likelihood information in the data.

The method of moments estimates parameters by equating the population moments with the sample moments and solving for the parameter(s) in question. The maximum likelihood estimate of a parameter is that value of the parameter which maximizes the likelihood function $L(\alpha, \beta)$ or the log likelihood $\ell(\alpha, \beta)$. Maximum likelihood makes more use of the probability model and generally yields more accurate estimates, especially with large samples. However, the method of moments can be useful for quick approximations or as a method for finding starting values for use in numerical maximization techniques to find the maximum likelihood estimate.